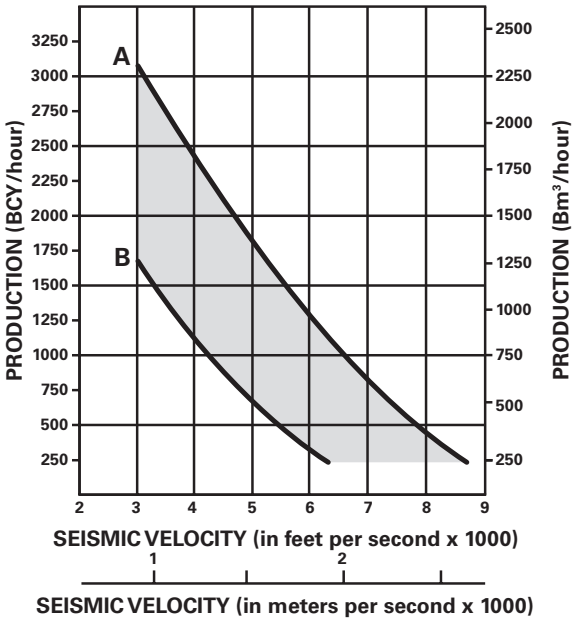
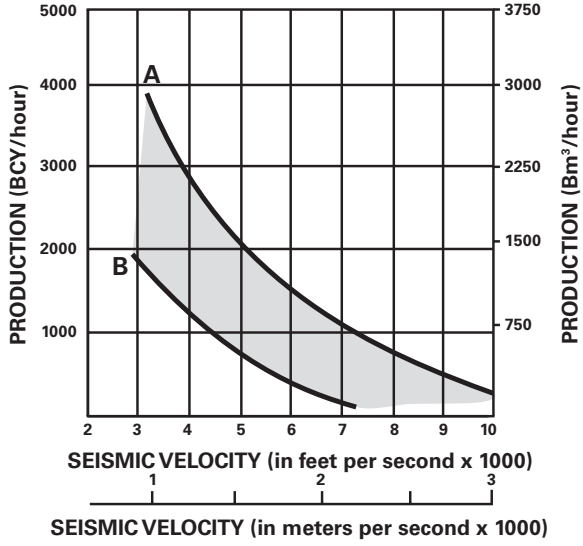
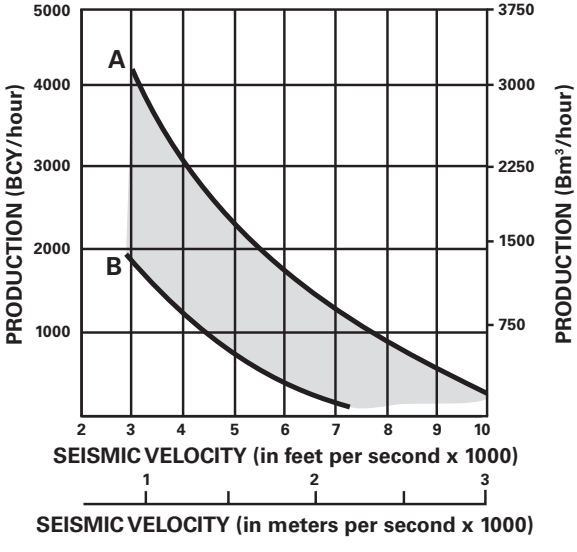
D10 WITH SINGLE SHANK



D11 WITH SINGLE SHANK



D11z CD WITH SINGLE SHANK



KEY
A — IDEAL
B — ADVERSE

HARD ROCK
Load-Haul-Dumps (LHDs)
Underground Trucks

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HARD ROCK

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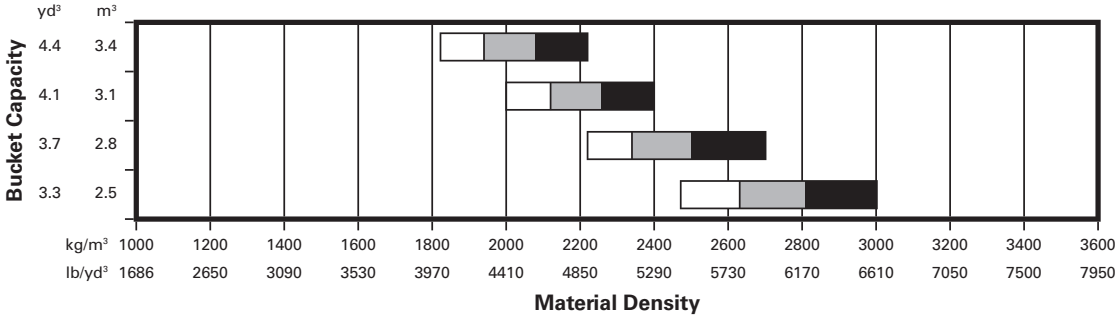
Features, all models:

- Rugged design for underground application.
- Engineered for productivity, reliability, safety and machine rebuildability.
- Extensive use of steel castings and forgings.
- Computerized machine function monitoring.
- Four wheel enclosed wet disc brakes.
- Remote Control Interface Group options and scalable autonomy options on loaders.
- Payload control system option on all loaders and selected trucks.
- Ride control system operational on all loaders and selected trucks.
- Fully enclosed air conditioned operator stations available.
- Operator Stations are ROPS certified.
- Autodig on LHD.
- Traction control.
- Operator safety monitoring.
- PLE data management.

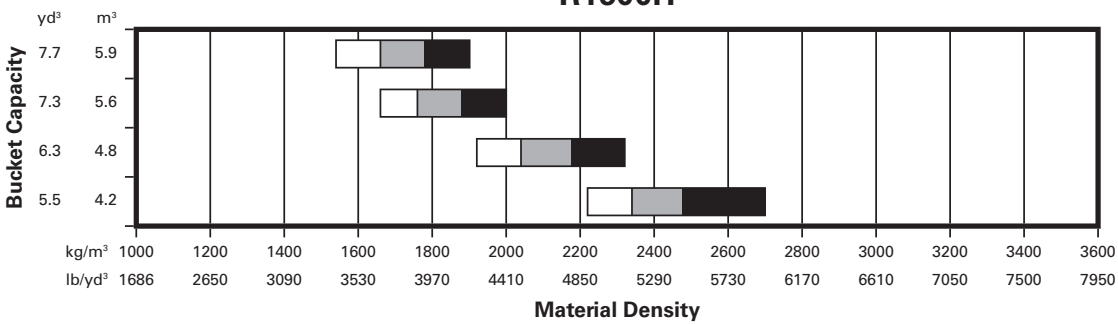
Product Line:

- Six models of Load-Haul-Dump (LHD) machines, with rated bucket payloads ranging from 6800 kg (14,991 lb) to 20 000 kg (44,100 lb).
- Five models of articulated dump trucks, with payload capacities of 22 000 kg (48,501 lb) to 63 000 kg (138,891 lb).

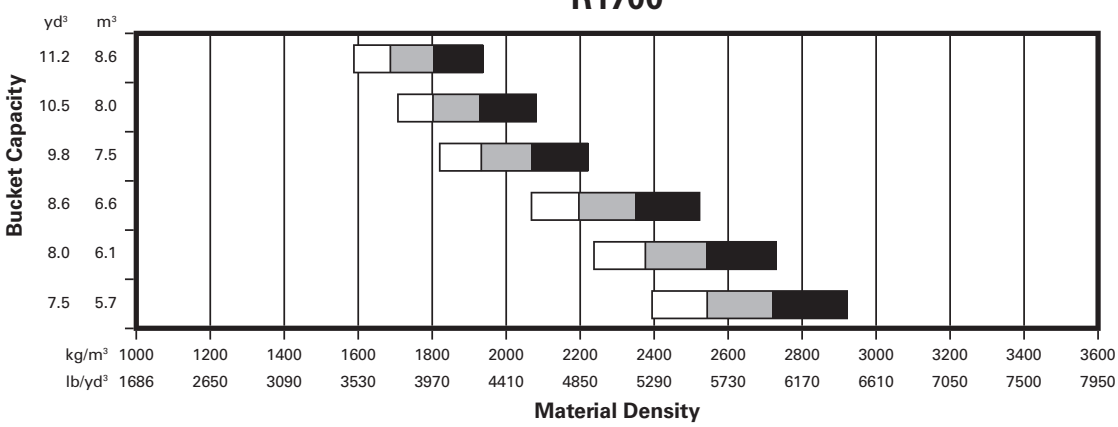
R1300G



R1600H

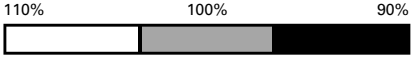


R1700

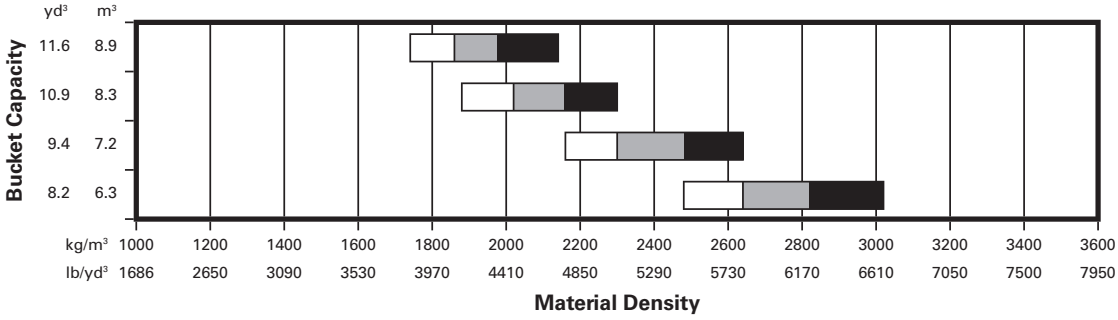


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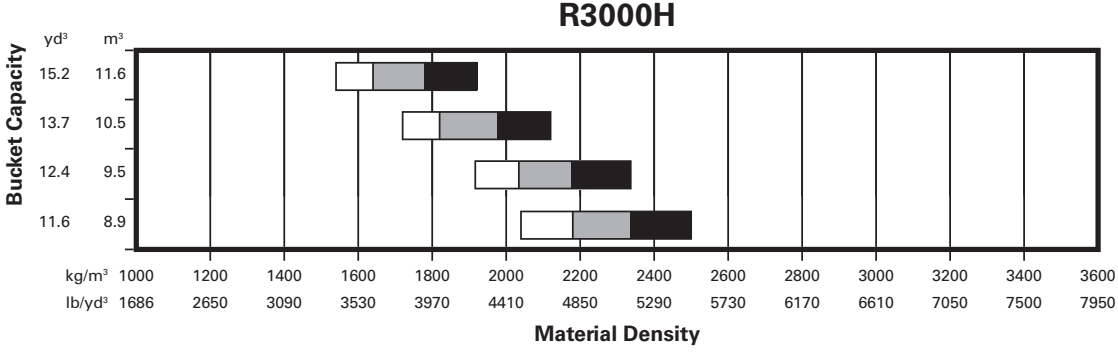
Bucket Fill Factor



R2900G

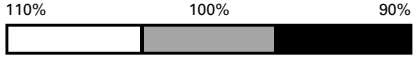


R3000H



KEY

Bucket Fill Factor



Turning Dimensions

Model	R1300G	R1600H	R1700	R2900G	R3000H
Outer Turning Radius	5717 mm **225.1"**	6638 mm **261.3"**	6857 mm **270"**	7323 mm **288.3"**	7536 mm **296.7"**
Inner Turning Radius	2825 mm **111.2"**	3291 mm **129.6"**	3139 mm **124"**	3383 mm **133.2"**	3247 mm **127.8"**
Articulation Angle	**±42.5°**	**±42.5°**	**±44.0°**	**±42.5°**	**±42.5°**

WASTE HANDLING

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INTRODUCTION

An increasing volume of refuse is generated by every person, household, commercial and industrial, entity day after day … 365 days a year. Disposal of this waste is a major concern worldwide. Increased governmental regulations designed to protect the environment along with rising transportation and land acquisition costs dictate that customers need to be cost conscience. Waste applications are very demanding and are a significant user of earthmoving and specialty mobile equipment, parts, and service. Caterpillar has Waste Specialists in each of the below categories. If you have questions, please contact your local dealer, Waste representative, or go to *Cat.com*.

SAFETY

Waste industry operators work in some of the world's harshest conditions, so Caterpillar has developed a range of safety resources to help your organization build a culture that delivers everyone SAFELY HOME. EVERYONE. EVERYDAY.™ Caterpillar Safety Services offers training programs, safety checklists, Toolbox Talks, videos, virtual walk arounds and other resources specific to the waste industry at *www.cat.com/safety*. Click on the Industry Safety tab, and then select Waste.

OVERVIEW

Waste applications are some of the harshest environments that machines and operators will work in. By definition waste is any discarded, rejected, abandoned, unwanted or surplus material.

The Waste Stream — It is important to understand the type of material you are working with when trying to spec a work tool and machine. Through many site assessments and machine productivity/competitive testing, we have found one underlying theme, the type of waste stream and site conditions, will dictate the type of work, tool, and machine. Residential waste (MSW), construction and demolition (C&D), green waste, commercial waste, wet waste*, industrials and sludges, auto fluff/car shred, etc., are just some of the types of waste that are dealt with daily. Each type of waste has different characteristics and properties. Weights of these materials could be as low as 148 kg/m^3 (250 lb/yd^3) to over 1187 kg/m^3 (2000 lb/yd^3) (sludges and soils). For example, during a recovering/recycling application, your work tool **Fill Factor** could be as high as **150–200%** when dealing with some MSW and shredded C&D.

MATERIAL RECOVERY FACILITIES, SORTING, AND TRANSFER STATIONS

Safety

MRF's, Sorting and Transfer Stations are not only harsh but busy environments. Inbound loads, mobile equipment, tip floor personnel, and outbound loads, are only a part of the traffic and dynamics dealt with minute by minute in very close settings. Proper PPE (personal protective equipment) for people, guarding for machines, and safety equipment on machines, are essential to keep safe.

Technology

Due to the demanding type of applications within MRF's, Sorting and Transfer Stations, machine technology has to lead the way in offering the ability to: scale loads, manage fuel and machine utilization, keep tire wear to a minimum, and help the operator with visual safety front and rear. Eco Modes on machines allow the ability to switch to a fuel saving mode. Product Link™ helps with tracking fuel burn, machine idle time and utilization. Traction control enables the operator to keep tire wear to a minimum.

Overview: Changing Attitudes in Waste Applications

MRF's (material recovery facilities) sorting, and transfer stations are becoming an important part of the waste industry. Diversion of waste from landfills is a global trend. The European Union, (EU), Canada, Japan, are just a few of the countries who divert 50–75% of their waste stream away from landfills. Cardboard, plastics, paper, food and green wastes are recovered, recycled, or composted. Major changes in the waste stream dictate different methods to gain density. The old mantra of "4–5 passes is enough," now becomes 4–7 passes so that materials can be shredded and bound together for higher densities. The U.S.A. is following this trend as regulations become stricter and the costs of siting/building landfills become higher. Transfer stations become a necessity as disposal sites continue to decline and/or move further away from city centers. MRF's, Sorting/Recycling and Transfer Stations all help reduce primary road and landfill traffic which helps create a safer environment for all.

*Wet Waste — Several country's, regions, and areas globally have concerns with waste that has a high moisture content. If you or your customer are working with a 'higher moisture content' type of waste, please see the *Wet Waste* section under *Landfills — Types of Waste and Refuse Densities*.

Primary Roles

- MRF's (Material Recovery Facilities) Sorting and Recycling Stations are designed to divert recyclable materials from the waste stream. Depending on whether the material is single stream, co-mingled, or pre-sorted, the goal is to divert recyclable material for re-consumption thus lowering the amount of waste placed in a landfill. The natural life of the landfill will be extended by this action. Additional savings of hauling costs can be realized by recovering/recycling materials close to the source (at the MRF or Transfer Station).
- Transfer stations, are designed to consolidate the loads of several residential vehicles in to long-haul vehicles such as rail cars, transfer trailers, barges, and ISO containers. On an average 2–5 inbound loads or more (depending on method of haul) can be consolidated into one outbound load. As a result, transfer stations offer cost savings over direct haul to landfills.

MRF's (Material Recovery Facilities), Sorting Stations, Recycling Stations

A well-coordinated recycling program will reduce volume going to the landfill by 50% or more. Trends are towards sorting and 1 of 2 types of MRF's.

- Dirty MRF — A dirty MRF accepts mixed waste, (normal household waste that has not been separated) that later may be sorted for recycling. Recoverable/recyclable materials are sorted either by hand or automation or a combination of the two. The materials are either baled and/or are hauled to a plant/industry that will utilize the recovered goods.
- Clean MRF — A clean MRF accepts source separated materials from residential or commercial sources. These are normally delivered in separate collection vehicles. As in the 'dirty' MRF, materials are either sorted by hand, automation, or a combination of both, baled and/or are taken to someplace that will recycle or utilize the recovered goods.
- Sorting Stations — Some transfer stations are designed for some waste diversion from mixed streams and are normally described as 'sorting stations.'

MRF's, Sorting and Transfer Stations Applications

There is a wide range of variability in MRF, Sorting and Transfer Station applications and operations. Depending on facility size, amount of material handled, type of station, etc. a wide variety of mobile equipment could be utilized.

In all the facilities mentioned above, materials are tipped by residential or commercial vehicles on to a tip platform or tip floor area. The materials are stockpiled then loaded into hoppers, conveyors, or a haul vehicle. Wheel loaders and excavators (tracked or wheeled) are the primary equipment for stockpiling, loading, sorting, topping off loads, and 'tamping' material into the haul vehicles.

"Volumetric consolidation of material" (normally in Transfer Station applications) is a method of compression and shredding that helps reduce the size of the waste loaded into a haul vehicle. It helps maximize the allowable weight carried by each haul unit. Common machines used for material consolidation are, track-type tractors, track loaders and compactors. These machines are equipped with track shoes or tips/teeth that help in shredding, compression, and reduction of the volume of waste. In most instances, machines performing the material consolidation/reduction are aided by a wheel loader or excavator.

For all applications within MRF's, Sorting, Recycling, and Transfer Stations, there is normally support equipment aiding the primary equipment. Small and compact loaders and excavators, backhoe loaders, skid steer loaders, and fork lifts, are just a few of these machines being utilized.

Types of Transfer Stations

Globally, top loading and compaction transfer stations are the two most popular designs. There are many different types of transfer stations, but, most conform to these two designs.

Top loading systems are the most common and simplest to operate. In top loading designs, there are normally five types of loading arrangements: direct, surge pit, full separation, half separation, and same level or floor loading. Depending on the type of transfer station, wheel loaders, excavators (tracked or wheeled), dozers, track loaders, and compactors can be utilized alone or in systems providing quick efficient waste handling.

Compaction transfer stations utilize hydraulic compaction equipment prior to or during the loading of the haul vehicle to compress the material into bales/logs or into the haul vehicle itself. The hydraulic compactor can be top loaded or side loaded by wheel loaders or excavators, depending on the type of design.

EQUIPMENT SELECTION

In waste applications, the largest single cost in daily operations is purchasing, operating, and maintaining the equipment that will handle the waste. Undersized, inadequate, or unreliable equipment results in inefficient operation and higher maintenance and fuel costs. As in many Cat® applications/industries, right sizing equipment should follow the familiar path of; Task, Tool, Machine. TASK: Understand the task or application the machine will work in. TOOL: What tool or tools will aid the machine in performing the task most efficiently? MACHINE: After finding the 'right tool(s)' to perform the job, then match it to the right sized machine.

Equipment in MRF, Sorting and Transfer Station applications perform the below operations.

1. **Stockpiling the Waste.** Once it is dumped onto the tip floor, wheel loaders, wheeled or tracked excavators, track-type tractors, track loaders, or compactors will push the material up into a stockpile. Stockpiling helps: recover storage and working space within the building, get the waste slightly denser, store the waste till it can be processed or loaded.

   Factors for a machine in stockpiling are: type of tool, reach, lift, ceiling height, storage space, and maneuverability.
2. **Recovering and Sorting.** Waste streams are rich in recoverable/recyclable materials. Waste diversion or recovery not only helps reduce the amount of waste going to landfills, but adds to profitability when sold to commodity markets. Sorting and recovery prior to stockpiling or loading can be accomplished with specialized machines equipped with sorting grabs, grapple buckets, or grapples. Stockpiled materials are sorted prior to loading or when loaded with grapples or grapple buckets. All sizes of equipment could be used depending on type of material and material flow. Compact wheel loaders/excavators, skid steer loaders, backhoe loaders, medium wheel loaders and excavators, and telehandlers should be sized according to amount of waste and type of application and type of tool(s) needed. It should be noted that in this type of application couplers on primary and support equipment could be a necessity.
3. **Loading the Waste.** No matter what type of haul/load out, vehicle/conveyor or hopper, materials are either pushed, load and carried, grappled or bucket loaded into transport vehicles or containers. Wheel loaders, tracked or wheeled excavators, track-type tractors, track loaders, compactors or any combination of these (systems) can be utilized. Note that the type of loading application will dictate the type of tool and machine necessary for optimum efficiency in the loading process.
4. **Reducing Volume of Material.** Haul vehicles/transport vehicles need to leave with the maximum allowable weight for the type of transportation mode being used. Wheel loaders and tracked or wheeled excavators can keep stockpiling or turning the waste to get the waste more dense. However, best compression, shredding and volume reduction of the waste is seen by layering waste in thin layers, continuous running over (passes) the waste, turning the waste, and "working" the waste. Track-type tractors, track loaders, and compactors normally see an average of about 60–120 kg/m^3 (100–200 lb/yd^3) higher density than wheel loaders or excavators in this type of application. Type of track or wheel/tip design is very important for shredding and compression in this type of application.

5. **Support Equipment.** Primary equipment often are supported by machines performing; clean up and sorting with specialized work tools, sweeping with brooms, stacking and loading with pallet forks or bale clamps, etc. Compact wheel loaders and excavators, backhoe loaders, skid steer loaders, and telehandlers equipped specifically for the tasks they need to perform can increase production and efficiency in all areas. Most of the time one or two machines are equipped with couplers to increase their versatility and efficiency helping lower costs.

Machine Selection Factors

Parameters that could influence tool and machine size, quantity, and possible combinations of machines in MRF's, Sorting and Transfer Stations could include:

1. Amount and type of daily waste to be handled (daily tonnage). During daily inbound tonnage, some peak delivery times occur. It has been found that approximately 50%–60% of inbound tonnages occur on 2–3 "Peak" hours/times of the day. Depending on the inbound tonnage during these times, machines could be spec'd towards these peak times (type and size of facility). What loading platform (top loading/compaction facilities) are they using and how much overall space to work in.
2. Facility load out dynamics.
   a. Maneuverability
   b. Ceiling height
   c. Floor size/storage space
   d. Floor arrangements
   e. Type of load out (below grade, ½ separation, same level, surge pit — loading)
   f. Haul unit specifications
   g. Inbound/outbound traffic patterns
3. Requirements for volume reduction for higher tonnages in load out.
4. Supplemental tasks performed in daily operations.
5. Budget.
6. Future growth.

NOTE: Waste applications place high demands on machinery requiring protection of the machine and its components. Special guarding is needed on all moving parts like axles and drive shafts. Windshields, fuel and hydraulic tanks, radiators, hoses, engine enclosures, are all vulnerable to debris and damage and require guarding. Specially designed debris screens and radiators will ensure clean out and better heat transfer for more efficiency.

Wheel Loaders

Wheel loaders are normally the primary machine used in MRF's, Sorting and Transfer Stations. Wheel loaders are designed, guarded and built in factory to withstand these harsh environments. Primary operations include: sorting and recovery of materials, push and stockpiling of waste, cleaning floors, support work, and loading conveyors/hoppers/and outbound haul vehicles. Depending on type of material, reach needed, amount of sorting required, ceiling height, floor wear, and type of load out, the machines can be equipped with a wide variety of buckets, attachments and tires. Keeping the thought process of Task, Tool, Machine in mind, a good knowledge of the facility, its waste stream, and type of support work, is needed to 'right size' the wheel loader(s) for these applications.

- Compact Wheel Loaders (906–920) — Normally used in small tonnage MRF's/Sorting Stations or as support in larger stations, maneuverability and versatility (coupler equipped) are its main strengths.
- Small Wheel Loaders (924–938) — Used in small–medium MRF's, Sorting and Transfer Stations and as support in larger stations, the small wheel loader brings maneuverability and versatility (coupler equipped) to these types of applications.
- Medium Wheel Loaders (950, 962, 966, 972, 980, 966XE and 972 XE) — When larger tasks, better reach, higher stockpiling, larger tonnages, etc. are needed, a medium sized wheel loader is normally utilized. These machines have excellent weight to horsepower ratio for better traction and moving larger loads.

The following Wheel Loader Operating Recommendations Chart is an estimate of possible machine(s) for tonnages and type of facility. (Note: Keep in mind the task/application the machine will work in, what tool(s) might accomplish the tasks and then size the machine.)

Wheel Loader Operating Recommendations

Tons per Day Volume	Clean MRF Recovery/Recycling	Dirty MRF/ Sorting Station	Waste Transfer Station	C&D MRF	C&D Transfer Station
0-100	906-930	924-930	924-930	930-950	930-950[b]
100-350	908-930[a]	924-930[a]	924-938[a]	950-966[d]*	950-966[d]*
350-500	930-950	930-950*	938-966*	966-980[a]*	950-966[d]*
500-1000	930-950[a]*	938-950[e]*	950-966[c]*	966-980*	950-980[d]*
1000-1500	938-950[e]*	938-966[e]*	966-980[d]*	950-980[d,e]*	966-980[d,e]*
1500-2000	950-966[e]*	966-980[e]*	966-980[d,e]*	950-980[e]*	966-980[d,e]*
2000-2500	950-966[e]*	966-980[d,e]*	966-980[d,e]*	950-980[e]*	966-980[d,e]*
2500-3000	950-966[e]*	966-980[e]*	966-980[e]*	950-980[e]*	966-980[d]*
3000 Plus	966-980[e]*	980[e]	980[e]	980[e]	980[d,e]

C&D = Construction and Demolition

*Depending on inbound/outbound tonnage, reach, production required, this machine could step up to either a 962 or 972.

a = Multiple machines recommended (contact your local Cat dealership for recommendations)
b = Depends on type and density of C&D material
c = Waste stream mixture may require multiple machines
d = Operating hours may require additional machines
e = Multiple machines required

The above are rule of thumb estimates. Your application might vary enough to change the machine for the above recommendations.
Other questions that might need answered are:

What are you loading? Conveyors, haul vehicles, hoppers, etc.?
If loading conveyors or drums — what are the feed capacities, FPS rate?
How much storage capability in your building?
What type of loading are you performing?
How much reach is required for stockpiling and loading?
"Other" required operations that the machine might perform?

Additional Considerations

All facilities are different and require special considerations when ordering and specifying tools and equipment. If tonnage and operating hours change, then tool and machine might change. Machine weight has to balance power to ground for best traction on slick floors. Keep in mind Task, Tool, Machine. (What type of work will the machine be asked to perform during the shift? What tool(s) will apply to that type of work cycle? What machine fits to that type of tool?)

It should be noted that most of the wheel loaders being sold into MRF, Sorting and Transfer Stations are high lift. This arrangement offers increased reach and lift height not only for loading and stockpiling, but in case the haul vehicle has to be brought to the same level as the machine to be loaded.

Hydraulic Excavators (Tracked and Wheeled)

Hydraulic excavators (tracked and wheeled) are often found in MRF, Sorting and Transfer stations either as a primary or secondary tool for loading haul vehicles/conveyors/hoppers, tamping and finishing off loads, and sorting materials. Fast cycle times are the primary advantage of an Excavator over a Wheel Loader. Wheeled excavators come with a maneuverability advantage especially in tight loading and stockpiling areas. Depending on the type of application, tracked and wheeled excavators would be equipped either for high production loading/compaction, or with sorting/separating tools for recovery of recyclable materials.

Again, keeping the Task, Tool, Machine concept in mind, you should take into account the following when sizing a tracked or wheeled excavator. Type of material and density, tonnage, work load for the machine, ceiling height, work area/swing area, floor size, along with maneuverability needed, ability to 'see' into haul vehicles or down into the floor load out area (cab risers), and reach. Special guarding packages, radiators, and reversing fans are options that may be needed to meet the most demanding needs.

Track-Type Tractors

Waste Handling Track-Type Tractors are designed, guarded, and built in factory so that they can accommodate all waste applications. Although its primary use is for pushing and layering the waste and cover materials in landfills they offer alternative waste movement and volume reduction in large transfer stations and C&D transfer stations. Ex: In surge pit operations where waste is tipped into a level below the tipping floor. Track-type tractors will push, layer, and compress/shred the material during operations and push into a haul vehicle. This method is used when maximum volume reduction is required or peak rate of waste exceeds available floor space.

Track Loaders

Waste Handling Track Loaders are designed, guarded, and built in factory so that they can accommodate all waste applications. Transfer stations are a strength for this machine due to its versatility in this application. Sorting, pushing, layering, and compaction are all part of typical operations within the transfer station application. Just like the track-type tractor, the track loader is used in higher tonnage transfer stations, C&D transfer stations, and when waste needs volume reduction. Equipped with a multi-purpose bucket, the track loader can sort, push, layer, compress and shred, and load material into outbound haul vehicles.

Landfill Compactors

Although landfill compactors are designed for spreading and compacting large volumes of material in a landfill environment, in some instances, transfer stations might have a large enough working area to allow the compactor to be utilized to reduce waste volume prior to loading into haul vehicles. They offer two advantages in this type of application.

1. They are configured and guarded to work in this type of environment.
2. They achieve higher compression, shredding, and compaction levels compared to other machines.

CAUTION! Operating a landfill compactor, track-type tractor, or track loader on a concrete floor could be counterproductive due to floor wear and maintenance costs. Always keep a layer of waste between the floor and machine until final movement of the day. Wheel loaders and/or excavators (wheeled/tracked) should be used to support these machines. (Please see #4 under previous Equipment Selection.) (It is not recommended to use a landfill compactor larger than the 826.)

Work Tools for Material Recovery Facilities, Sorting and Transfer Stations

As described in the opening statements of this section, the waste stream can consist of many different types of materials sometimes all blended together. For that reason, work tools become a very important part of moving, sorting, and dozing in waste applications.

At the end of this section, there will be a Work Tool section and chart discussing possible tools to use. Please contact your dealer/regional waste representative for more information.

LANDFILLS

Safety

Landfills are harsh and busy environments. A variety of vehicle traffic on haul roads, busy tipping areas, people on the tip floor, a large variety of different types of waste, mobile heavy equipment operations, special waste haulers, etc. are all part of the dynamics of a landfill working face. Proper PPE (personal protective equipment) for people, guarding for machines, proper training, and safety equipment on machines, are essential for a safe work environment.

Compaction Power with the Cat Compaction Technology

Our Accugrade™ Cat Compaction Technology software supplies dependable, real-time data in the form of pass count, compaction and cut/fill mapping, as well as map sharing. Utilizing 3D mapping technology, Cat Compaction Technology offers system options for a variety of landfill sizes and waste management needs. The result? With proper machine usage, you could see up to 40% better compaction. Ask your dealer for more details.

Payload Systems for Loaders, Haulers and Managers

Whether you're using wheel loaders or excavators, on-highway trucks, off-road haulers or hoppers, there is Cat Payload technology made for your applications. Operators can see how much they're lifting in every bucket load and when they've reached optimum hauler capacity. Managers can then track tonnage output on whatever time frame they need.

Cat® Command is revolutionizing equipment operation. By removing the operator from the machine, Command opens up new possibilities—for safe operation in hazardous areas, for improved sightlines in tricky applications, for enhanced productivity by putting operators in more comfortable settings or by enabling them to manage multiple machines at once.

Cat® Detect safety technology helps operators become more aware of their surroundings and automatically prevents them from engaging in certain unsafe operations. Proximity sensing systems limit machine movement in tight spaces, while cameras and other sensors keep operators informed about equipment and personnel working near their machines.

Cat® Inspect allows you to access your equipment data on your mobile device. This easy-to-use app lets you capture inspection data and integrates with your other Cat data systems, so you can keep a close eye on your fleet. More than one million inspections are completed each year, providing convenience and accountability to equipment owners.

Technology

Due to the demanding environment and its possible applications, machine technology has to lead the way in offering the ability to: manage fuel and machine utilization, keep undercarriage cost to a minimum, and help the operator with visual safety front and rear. Eco Modes on machines allow the ability to switch to a fuel saving mode. Product Link helps with tracking fuel burn, machine idle time and utilization. Computers, VIMS™ (Vital Information Management Systems,) and Traction Control (in some track-type tractors) helps operators control spin and keep track of mileage to help decrease undercarriage wear.

Landfill Overview: Landfilling Now and in the Future

Although recovering and recycling materials is becoming more prevalent, landfilling still provides a place to deposit waste materials or refuse not able to be recovered/recycled. A landfill protects the environment by disposing of the waste in an engineered cell. Siting, designing, building, and operating a landfill is costly. Depending on country, state, region, etc., landfills have to design and operate to very strict regulations. Proper equipment selection and daily operating techniques can maximize and extend the operational life of the landfill along with lowering costs.

Primary types of landfills are MSW (municipal solid waste/residential), C&D (construction and demolition), and Hazardous (hazardous industrial sludges, asbestos, etc.) with the majority of landfills being MSW or residential waste. Landfills can take in many different types of materials and will range from less than 90 metric tons/day (100 tons/day) to over 13 610 metric tons/day (15,000 tons/day). Depending on the country you live in and its regulations on recycling, the waste stream going to the landfill could be a mix of food wastes, packaging, plastics, cardboard/paper, metals, C&D, etc. In some highly regulated countries, food wastes, plastics, cardboard/paper, metals, and C&D, are diverted to C&D landfills, composting sites, and/or facilities that will recycle the recovered product.

In keeping with environmental concerns, most landfills have a highly engineered method of dealing with leachate, gas, and inbound waste. (See the *Caterpillar Equipment and Application Guide — Waste Landfills* or your countries Environmental Landfill sites.) Once cell, leachate, and gas methane extraction development are in place, the basics are to: push, layer, compact the waste in thin layers, adding daily cover or ADC (alternative daily cover) sparingly. The idea is to fill the landfill with as much waste as possible without robbing airspace by using too much cover soil.

Diversion rates of recoverable materials within the EU Canada, some Asia Pacific countries, and starting in the U.S. will eventually reach 60%–80% diversion or more. This will all depend upon commodity markets and the demand for the recycled materials. It also drastically changes the waste stream going to the landfill. The type of waste going to a landfill from a good diversion plan is normally light plastics, packaging, hard to handle materials, metals and materials that can't be recycled. This waste stream might require different handling, processing, amount of passes, and compaction techniques to shred and compress it as much as possible and possibly different wheel tips and tip patterns. (See text in *Factors Governing Compaction —Developing a Pattern* and *Running Proper Passes*.)

EQUIPMENT SELECTION

A landfill's largest single cost for daily operations is purchasing, operating, and maintaining its mobile equipment fleet. Undersized, inadequate or unreliable equipment results in low machine utilization, higher fuel and maintenance costs and eventually improper landfill operations.

Landfill equipment performs three major distinct functions:

1. Waste disposal: Which includes, pushing, spreading, and compaction. Track-type tractors, track loaders, and landfill compactors are primary machines in this application.
2. Cover material mining and application: Machines will provide daily cover requirements whether cover soil or ADC (alternative daily cover). If supplying cover material is a machine's sole function, then it can be selected on the same criteria as normal earthmoving. Distance to borrow pit, material characteristics, volume to transport, production needs, underfoot conditions, etc. are all important in right sizing the equipment. Lowest cost is determined by best efficiency and lowest cost/m^3 (yd^3).
3. Support activities: What 'other' roles might the equipment be asked to accomplish or what support is needed for the primary equipment? Water trucks to keep dust down, motor graders to maintain haul roads, wheel loaders/excavators/backhoes/skid steer loaders, telehandlers, etc., with couplers that can utilize all types of work tools, generators, service vehicles, air compressors, etc. are all necessary equipment to keep the landfill running efficiently.
4. Peak periods: Most landfills have to deal with 2–4 periods during the day where more volume of waste is delivered by more inbound vehicles than normal. These are called peak periods. Depending on the size of the landfill and amount of peak waste coming in, it should be noted that some landfill equipment can or should be specified in order to handle these peak times of waste and vehicles.

There are many variables when choosing the "right" sized equipment for landfill use. Again, we point to the direction of thinking: Task, Tool, and Machine. What are the application, production, compaction, support requirements? What tool or tools can accomplish this task? And finally, what machine(s) will handle the tool(s) most efficiently? (To help with 'right sizing' a compactor and/or track-type tractor together, Waste Fleet Analysis/WFA (downloaded at *Dealer.Cat.com*) can be utilized as a rule of thumb to determine possible compactors/track-type tractors and combinations of the two.)

Track-Type Tractors

Waste Handling Track-Type Tractors are designed, guarded, and built in factory so that they can accommodate all waste applications. The track-type tractor is the most popular machine on the landfill. They prepare the site, build haul roads, push/spread/and sometimes compact the refuse, spread cover material, and perform a variety of support functions. Quantity of material to be moved/hr., type of inbound delivery vehicles, distance of push, support work needed, etc., all should be taken into consideration when sizing a track-type tractor for its application. Economic distances for dozing waste or cover soil efficiently is no more than 90 m (300 ft) or less. The track-type tractor, when used to compact, can achieve densities of approximately 475–590 kg/m^3 (800–1000 lb/yd^3) (MSW, 3:1 or less slope, multiple passes).

Track Loaders

Waste Handling Track Loaders are designed, guarded, and built in factory so that they can accommodate all waste applications. These machines are highly versatile and perform many primary and support activities. They can be used in any size landfill and are ideal for landfills under 135 metric tons (150 tons) since they can work solo to perform waste handling and cover material functions. Equipping the machine with a coupler or a multi-purpose bucket increases its versatility. The multi-purpose bucket can be used for pushing waste, sorting materials, and dozing cover soils. The machine can be equipped with different track options from double to single grouser allowing it to work in normal operations or be utilized to more aggressively shred materials (Heavy MSW or C&D). Utilizing correct methods of compaction, (spread thin and load the bucket with soils to increase weight) will achieve higher densities during compaction. Compaction densities can range from 475–590 kg/m^3 (800–1000 lb/yd^3).

Landfill Compactors

Landfill Compactors are specifically designed, built, and guarded in the factory to effectively shred and compress large volumes of waste. Depending on landfill size and amount of tons inbound daily, the compactor will work solo (pushing, spreading, compacting) handling all the inbound waste, or work in tandem with a track-type tractor (track-type tractor — push/spread, compactor — shred/compact). When working the track-type tractor and landfill compactor together as a "system," productivity and compaction densities will increase. Depending on the type of waste stream to be compacted, initial waste density, depth of the layer spread, and size/weight of the landfill compactor, compaction densities can be 593–1100 kg/m^3 (1000–1854 lb/yd^3). Although the compactor can work on steeper slopes, best slopes for highest density for landfill compactors are 4:1 or less.

Wheel Loaders

Wheel loaders are designed, guarded and built in factory to withstand these harsh environments. Not recommended as a compaction machine, wheel loaders are mobile and versatile and sometimes are used by communities sharing a single machine which can travel and support community/landfill projects. Wheel loaders are normally equipped with a coupler and many different tools to make it more versatile. Loading cover soils, sorting, clean up, spreading road material, and handling leachate/gas pipes are just a few of the activities wheel loaders perform in landfill applications.

Hydraulic Excavators (Tracked/Wheeled)

Every landfill has a tracked or wheeled excavator supporting its operations. Whether equipped with a coupler, which accommodates multiple tools, or a pinned attachment, wheeled and tracked excavators offer superior versatility in: cell development, cover soil mining, ditch/pond clean out, and leachate and gas extraction support. Wheeled excavators give you mobility and maneuverability along with the ability to be able to respond quickly to support needs. When loading cover materials is the primary application, it is critical to know the size of the haul vehicle, type of material, and production required, so as to define the proper size bucket, stick, and finally machine to meet production needs.

Wheel Tractor-Scrapers

Wheel tractor-scrapers mainly perform cover soil operations. Wheel tractor-scrapers work in site preparation, cell construction and hauling/placing cover material. When underfoot conditions are good and the haul over 185 m (600 ft) the wheel tractor-scraper will be economical. The machine should be selected as if performing a typical earthmoving application. Average sized landfills prefer the work alone capability of either an elevating or twin engine scraper since they are self-loading machines. Larger landfills sometimes utilize larger push–pull scraper units to meet their daily cover soil production needs. Preferably, the scraper unloads the cover material close to the working face where track-type tractors or track loaders can push and spread the material in the required layer depth. This will reduce tire damage and extra maintenance costs to the machine.

Articulated Trucks

Articulated trucks are versatile, highly maneuverable, all weather haulers that can negotiate poor underfoot conditions. Like wheel tractor-scrapers, articulated trucks work in site preparation, cell construction, and hauling/placing cover material. Articulated trucks are normally loaded by a variety of loading tools and economically effective with hauls from 0.2 km–5 km (600 ft–3 miles). In dump configuration, cover material is dumped close to the work face and spread by a supporting track-type tractor or track loader. Ejector configured trucks are preferred and provide safer on-the-go dumping in normal to softer more sloped ground conditions where a "dump configuration" machine would not be suitable. Cat articulated trucks offer optional container handling and refuse body configurations for specialized landfill applications.

Work Tools for Working in Waste Landfill Applications

As described in the opening statements of this section, the waste stream can consist of many different types of materials sometimes all blended together. For that reason, work tools become a very important part of moving, sorting, and dozing in waste applications.

At the end of this section, there will be a Work Tool section and chart discussing possible tools to use. Please contact your dealer/regional waste representative for more information.

Machine Selection Factors

Selecting the type, size, quantity, and combination of machines required to spread, compact, and cover varying daily refuse volumes is determined by the following parameters:

1. Amount and type of waste to be handled (daily tonnage) (peak periods should be charted and reviewed and taken into considerations)
2. Amount and type of cover material to be handled
3. Distance cover material to be transported
4. Compaction/density requirements
5. Weather conditions
6. Supplemental tasks
7. Budget
8. Growth

A. Daily tonnage and peak periods — Amount of waste produced by a community is the major variable in selecting the appropriate sized machine. The following chart serves as a "guideline" in sizing a landfill machine. For example, if a community generates approximately 180 metric tons (200 tons) of refuse per day, a D6, 953, and/or an 816F2 landfill compactor could be considered. Depending on the tonnage, type of waste, and peak periods, multiple machines or a 'system' of machines (ex: 836K and D8T) could be considered. As discussed earlier, peak periods during the day, (periods of time where a higher than normal average amount of waste comes in) could dictate what size machine is needed to keep up with the inbound waste. During these times, another machine could be added to the mix until the waste is layered and compacted correctly.

B. Amount and type of cover material to be handled — Landfill, size, type, and methods of operation vary dramatically from site to site. The type and amount of cover material utilized is important. The use of Alternative Daily Cover (ADC) is highly recommended to reduce loss of valuable airspace. Landfill managers track cover material use as close as they track their airspace and maintenance. The use of cover material is broken down into three segments: daily, intermediate, and final. Landfill operators agree that daily cover should be no more than 10% or less of the entire lift. No matter how much daily or intermediate cover is used, the growing trend is to remove or 'mine' cover material prior to beginning the daily operation in that area. The total combination of daily and intermediate cover if not removed, could end up being more than 25%–35% of total landfill airspace loss. Removal of cover material increases airspace and helps facilitate leachate and gas migration.

   Blade/bucket design is paramount if a machine is used specifically for working with cover soil. Straight blades, Semi U blades, and multi-purpose buckets are primarily used by track-type tractors and track loaders. With the need to carry and layer cover material to an exacting maximum, blades/buckets that can shed material easily are recommended. If the machine will be used in waste and cover soil operations, then a Semi Universal blade (for load and carry) and the multi-purpose bucket are good recommendations.

C. Distance cover material is to be transported — Whether working with daily, intermediate, or final cover, the following guidelines are recommended for cover material movement. Quantity of material to be moved, required production, and maintenance results, must also be considered when using these guidelines.

Track-type tractor	0-61 m (0-200 ft)
Track loader	
Push/spread	0-61 m (0-200 ft)
Load & Carry/spread	0-150 m (0-500 ft)
(Type of material/application needs to be considered for possible excessive track wear)	
Wheel loader	0-185 m (0-600 ft)
Wheel tractor-scraper	over 185 m (over 600 ft)
Articulated truck	over 185 m (over 600 ft)
(Haul distance, road and borrow pit conditions, weather, tip area dynamics — all need to be taken into consideration when making a decision to use a scraper or articulated truck)	

D. Compaction requirements — Best utilization of airspace is critical to extending landfill life. Quantity and type of material, work face variables, operator skill, operating hours, size and type of equipment, etc., all play a part in achieving high densities. Highest density is gained by utilizing a 'system' approach of machines (track-type tractor pushing and spreading — compactor leveling and compacting) or to a lesser extent, just using a compactor.

The following product segments of this section will contain features, specifications, and work tools for primary Cat Waste Handling machines. Additional drawbar/rimpull vs. groundspeed charts, ground pressures, controls, production estimating tools, information, can be found in respective product sections within this *Performance Handbook*.

The below *Machine Tonnage and Usage Selection Factor Guide* should be considered a 'rule of thumb.'

To use; find the amount of tonnage/day that the landfill takes in (left column) — to the right of that column find the possible machines that could be used for that amount of tonnage (marked by an ×). Please note that depending on the type of waste, amount of waste/day, and even amount of waste/hr., a single machine could be utilized or multiple machines might be necessary.

For example: Follow the left column down to 45.3 to 136 Metric Tons/day (50 to 150 tons per day) — For MSW Landfills — All the machines marked by ×'s could be utilized. Smaller machines for smaller tonnages, larger machines for larger tonnages. Depending on budget and required density, and required support activity, a single machine might be used or a combination of machines could be needed.

Machine Tonnage and Usage Selection Guide	D5 WHA	D6 WHA D6XE WHA	D7 WHA	D8T	D9T	953	963	973	816	826K	836K
0 to 45.3 Metric Tons per Day (0 to 50 Tons per Day)											
MSW Landfills	×	×				×	×				
Construction and Demolition Landfills/Other		×	×				×	×			
45.3 to 136 Metric Tons per Day (50 to 150 Tons per Day)											
MSW Landfills	×	×				×	×		×		
Construction and Demolition Landfills/Other		×	×				×	×	×	×	
136 to 227 Metric Tons per Day (150 to 250 Tons per Day)											
MSW Landfills	×	×				×	×	×	×	×	
Construction and Demolition Landfills/Other		×	×	×			×	×	×	×	
227 to 317.5 Metric Tons per Day (250 to 350 Tons per Day)											
MSW Landfills		×	×	×			×	×	×	×	
Construction and Demolition Landfills/Other			×	×				×		×	
317.5 to 453.6 Metric Tons per Day (350 to 500 Tons per Day)											
MSW Landfills		×	×	×			×	×		×	×
Construction and Demolition Landfills			×	×	×			×		×	×
453.6 to 680.4 Metric Tons per Day (500 to 750 Tons per Day)											
MSW Landfills			×	×			×	×		×	×
Construction and Demolition Landfills/Other			×	×	×			×		×	×
680.4 to 907.2 Metric Tons per Day (750 to 1000 Tons per Day)											
MSW Landfills			×	×	×			×			×
Construction and Demolition Landfills/Other			×	×	×			×			×
907.2 to 2721 Metric Tons per Day (1000 to 3000 Tons per Day)*											
MSW Landfills			×	×	×			×			×
Construction and Demolition Landfills/Other			×	×	×			×			×
2721 PLUS Metric Tons per Day (3000 PLUS Tons per Day)*											
MSW Landfills			×	×	×			×			×
Construction and Demolition Landfills/Other				×	×			×			×

*Might require multiple machines or 'systems.' Systems = dozer/compactor combination (S).

TYPE OF WASTE AND REFUSE DENSITIES

The type and density of the waste stream inbound to MRF's, Sorting/Transfer Stations, and Landfills determine how it is handled, the type of work tool, and ultimately the type of machine to handle it. Type of waste to be handled will strongly influence machine selection. The major waste components for the community and landfill should be identified and machine selection based on the type of waste and compaction density required. For example, if the site receives a high proportion of material that is hard to compact like medium to heavy C&D waste, (rocks, bricks, concrete, tree stumps, telephone poles, etc.) a landfill compactor might have a hard time achieving normal densities without the help of a track-type tractor or track loader. Tracked machines have more difficulty pushing and compacting bulk waste such as trees, road materials, iron, etc. A combination or "system" of both tracked machine and landfill compactor might work best. Depending on the type of waste, different tip/cleat designs will help with shredding, traction, and density.

On average, Americans generate approximately 2 kg (4.4 lb) waste/day while the European Union (EU) generates about 1.4 kg (3.0 lb) waste/day. (Before recovery/recycling) Waste composition varies from location to location, the following charts are representative of the waste stream in the U.S. and EU. (Please note that the growing trend within the EU and to a lesser extent the U.S. is to use waste as a resource. Subsequently the numbers represented in these charts will have annual changes.) The EU-27 are avoiding waste to landfills through recovery, recycling, composting, and incineration, and are trying to use waste as a resource. For the EU-27, approximately 38% of waste is landfilled. The U.S. is currently approximately 33% recovered/recycled with a slow trend towards composting.

Generally, loose residential and commercial refuse weighs 150–267 kg/m^3 (250–450 lb/yd^3). A refuse truck will increase this density to 237–593 kg/m^3 (400–1000 lb/yd^3). Once ejected from the refuse truck, some waste has a 'rebound' tendency and it will lose some density. Typically, loose on the landfill, we see loose refuse weights with an average of 207–475 kg/m^3 (350–800 lb/yd^3). This waste needs to be compacted and in-place densities can vary from between 356–889 kg/m^3 (600–1500 lb/yd^3) depending on the shredding and compression applied by the compacting machine. C&D landfill sites, depending on type of inbound material, sometimes see a wide range of densities from 593–1187 kg/m^3 (1000–2000 lb/yd^3) (depending on compacting machine). Cover material will raise fill densities 119–296 kg/m^3 (200–500 lb/yd^3) from the numbers above. When asked about the density of their landfill, most managers will give you the 'in-place with cover' estimates of density — Please see weights of refuse below:

Weight of Refuse

	kg/m^3	lb/yd^3
Loose Refuse:	148-237	**250-400**
Packer Truck:	237-474	**400-800**
Fill Density:	355-949	**600-1600**
Refuse and Cover:	474-1186	**800-2000**

The Current U.S. Picture

EPA began collecting and reporting data on the generation and disposition of waste in the United States more than 35 years ago. The Agency uses this information to measure the success of materials management programs across the country and to characterize the national waste stream. These Facts and Figures are current through calendar year 2018.

The total generation of municipal solid waste (MSW) in 2018 was 292.4 million tons (U.S. short tons, unless specified) or 4.9 pounds per person per day. Of the MSW generated, approximately 69 million tons were recycled and 25 million tons were composted. Together, almost 94 million tons of MSW were recycled and composted, equivalent to a 32.1 percent recycling and composting rate. An additional 17.7 million tons of food were managed by other methods. Other food management includes the following management pathways: animal feed, bio-based materials/biochemical processing, co-digestion/anaerobic digestion, donation, land application and sewer/wastewater treatment. For more information on food management, see Food: Material-Specific Data. In addition, nearly 35 million tons of MSW (11.8 percent) were combusted with energy recovery and more than 146 million tons of MSW (50 percent) were landfilled.

More Information

Check out the Facts and Figures methodology.

See our Sustainable Materials Management web area for relevant information and our State Measurement Program page for state-specific information.

EPA refers to trash, or MSW, as various items consumers throw away after they are used. These items include bottles and corrugated boxes, food, grass clippings, sofas, computers, tires and refrigerators. However, MSW does not include everything that may be landfilled at the local level, such as construction and demolition (C&D) debris, municipal wastewater sludge, and other non-hazardous industrial wastes. While the analysis in Facts and Figures focuses primarily on MSW, EPA has been including estimates of C&D generation and management as a separate non-hazardous waste stream in recent years.

Recycling

The total MSW recycled was more than 69 million tons, with paper and paperboard accounting for approximately 67 percent of that amount. Metals comprised about 13 percent, while glass, plastic and wood made up between 4 and 5 percent.

Measured by tonnage, the most-recycled products and materials in 2018 were corrugated boxes (32.1 million tons), mixed nondurable paper products (8.8 million tons), newspapers/mechanical papers (3.3 million tons), lead-acid batteries (2.9 million tons), major appliances (3.1 million tons), wood packaging (3.1 million tons), glass containers (3 million tons), tires (2.6 million tons), mixed paper containers and packaging (1.8 million tons) and selected consumer electronics (1 million tons). Collectively, these products accounted for 90 percent of total MSW recycling in 2018.

Wet Waste

There are a number of global landfills that deal with a heavier than normal amount of 'wet waste,' or waste with higher than normal moisture content. In previous discussions, 'type of waste' will determine how the waste is handled or worked. With higher than normal moisture contents, the idea of 'multiple' passes (3–5) will be changed. For example, the waste will already be heavier/denser than normal (ex: 1186 kg/m^3 or 2000 lb/yd^3) this type of waste might only require a track-type tractor or track loader to spread it. During the spread cycle, the shredding/tearing action of the track cleats provides an additional density. Or, the compactor could be equipped with a 'wider' flotation wheel (optional wheel from Caterpillar) that provides better flotation along with shredding and compression that will add to the density of the waste. Whichever system is used, less passes on the waste are needed to reduce the waste and add to density. (Too many passes could lead to the waste breaking down and causing traction/flotation problems on the landfill.)

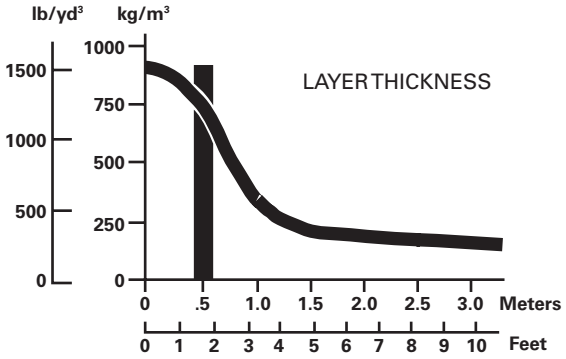
FACTORS GOVERNING COMPACTION

There are four factors that will determine best compaction density (layer thickness, patterns/passes, slope, and moisture content). The three areas that can be affected most by proper machine applications are:

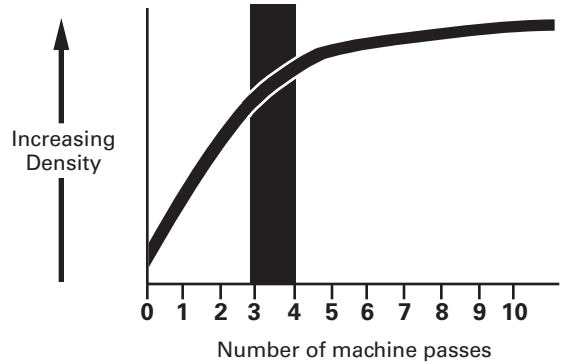
1. Refuse Layer Thickness — The single most important factor in gaining maximum density is the depth of the layer being spread to compact. Depending on the amount of tips/wheel, type of tips, type of waste, and weight of the machine, optimum layer depth will vary. Through field testing and computer analysis, optimum layer depths for highest density have been analyzed. Below are guidelines of optimum layer depth for each compactor.
   a. 816 — 0.3–0.4 m (12–15 in)
   b. 826 — 0.45–0.5 m (18–20 in)
   c. 836 — 0.5–0.6 m (20 24 in)

   Layers may be placed thicker however; density will be reduced no matter how many passes the compactor performs.

   Below is a Layer Thickness chart based on MSW and 4 passes by an 836.



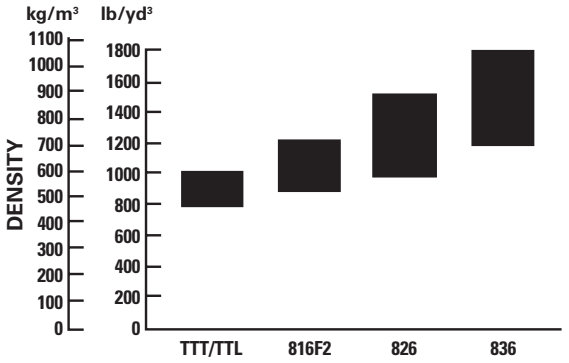
2. Developing a Pattern and Running Proper Passes — Maximum compaction density is obtained by putting the waste down in proper layer height, running a proper pattern and going over the waste with the proper number of passes. Those operators who develop a pattern, (starting from one side of the working face and running passes over entire area that waste has been layered in a logical sequence) and run proper amount of passes, normally obtain high density. In the past, it was said "that 3–5 passes is maximum to achieve highest density. Any more passes does not justify the added fuel and maintenance for the incremental gain in density." Again, through testing and computer modeling, it has been noted that running an extra 1–2 passes diagonally (45 degrees) over the waste after the first 4 passes, can achieve higher density through 'shredding and knitting' the waste together (less cover soil/ADC could also be used due to the blending of the waste). In many countries, the 'diversion' of compost and biodegradable materials leaves some waste streams very dry. Drier waste streams demand more passes than normal, (sometimes 5–8 passes,) so that the material can be shredded or broken down properly for best compaction density.



3. Slope — Maximum density is achieved by shredding and compression of material. Track-Type Tractors and Track Loaders achieve highest density on a 3:1 or less slope. The dynamics of track and gravity working together help shred material into smaller pieces. Although Landfill Compactors can work on steeper than 4:1 slopes, their best density is realized at 4:1 or less. The flatter the slope the better the density. The weight of the compactor is more efficiently utilized and concentrated on flatter slopes. Slight slopes sometimes add higher density due to the shearing stress that aids in shredding and blending of materials. Depending on degree of slope and whether working with another machine, (track-type tractor, dozer, or compactor) fuel use can sometimes be better uphill or downhill. Of course production is always better downhill. Again, depending on type of material and slope, tip selection for the compactor should be taken into consideration.

COMPACTION COMPARISON ESTIMATE

The following graph may be used as a rule of thumb for the compaction ranges of landfill machines assuming the proper operating technique is employed. (The mid ranges of each graph is assuming average waste stream and average operator.)



With correct operating procedures, tip/track/tool selection, and finally machine selection, density can be increased. An example of how increased density can increase landfill life is depicted in the next charts.

EXAMPLE OF INCREASED COMPACTION ON POTENTIAL LANDFILL LIFE

Landfill refuse capacity	1 530 000 m³ **(2,000,000 yd³)**
Operating days	260
Daily volume	365 metric tons **(400 tons)**
Yearly volume	94 328 metric tons **(104,000 tons)**

Compaction	Landfill Life	Gain
590 kg/m³ **1000 lb/yd³**	9.6 years	0
710 kg/m³ **1200 lb/yd³**	11.5 years	1.9 years
830 kg/m³ **1400 lb/yd³**	13.4 years	3.8 years
950 kg/m³ **1600 lb/yd³**	15.3 years	5.7 years
1070 kg/m³ **1800 lb/yd³**	17.2 years	7.6 years

COMPACTOR PRODUCTION GUIDELINES

Model	Metric Tons/hr	U.S. Tons/hr
836K	140	**135**
826K	120	**115**
816F2	65	**65**

Parameters for the above: MSW, 4 passes, 18–30 m **(60–100 ft)** push distance, 4:1–5:1 slope, good operator.

All models are making 3 to 4 passes.

A pass is defined as: A machine traveling over the refuse one time in one direction on flat ground.

Adverse (uphill) or favorable (downhill) grades, deep layered waste, hard to handle waste, etc., will affect the above production figures.

(For tons/day — multiply by the amount of hours the machine will work in a days time.)

A. Weather Conditions — Inclement weather affects production and density. Equipping machines with the proper track shoe or wheels and tips for the typical climate it works in will increase production and density while decreasing costs.
B. Supplemental Tasks — There are a wide variety of tasks to be performed daily on a landfill. Understanding the 'supplemental' tasks that a machine might be asked to perform daily is paramount to adding the work tool(s) and right sizing the machine. For example, will the machine be required to perform site clearing, asked to build or maintain access roads, excavate cover soil, etc.? Logical choices of machines could change according to type of tasks and versatility required.
C. Budget — Smaller landfill operations with limited budgets may have to consider single machine versatility ahead of specialized machines or multiple units.
D. Growth — Population growth or added contracts can both increase and change inbound refuse. Future increases/decreases in refuse, type of material, density of material, etc., must be considered to properly pick tools and size machines.

LANDFILL ESTIMATING

Example Problem #1

A professional engineer has developed a small, rural landfill master plan. The local legislative regulatory agency has approved the plan and site.

Assume:

Waste generation: 2.04 kg/day (4.5 lb/day) per person
Waste collection: 6 days/week
Topography: flat
Land availability:
area has several suitable sites at nominal price
Population served: 30,000
Projected population in 3 years: 40,000
Current daily refuse volume: ?
Type of refuse: mostly household, some commercial
Operation: propose 8 hours/day, 5½ days/week
Present equipment: none — new site

What would your comments and recommendations be on the following:?

a. Probable amount of refuse generated daily?
b. Type of machine for the proposed Landfill?
c. Size of machine for the proposed Landfill?

Solution

a. The current incoming waste stream can be determined to be 2.04 kg/day (4.5 lb/day) per person × 30,000 people = 61.2 metric tons (67.5 tons) daily. You must now multiply that daily generation rate by 7 for the total weekly generation, and divide by the number of days that the waste is collected (6). Therefore, your waste collection per day will be (61.2 tons/day × 7 days)/6 days of collection = 71.4 metric tons (78.7 tons) collected daily.

   The same equation can be used to determine the three year projected waste stream of 40,000 residents to be 81.6 metric tons (90 tons) generated daily, 95.2 metric tons (105 tons) collected each day.
b. Track loader — excavating ability, single machine application based on tonnage requirements.
c. 953C handle current refuse, and has extra capacity for future growth. Small compactor if additional compaction is required.

Example Problem #2

Existing landfill has been in operation for several years.

Assume:

Type of operation: area fill
Cover material: suitable material within 90 m (300 ft)
Current daily refuse volume: 500 metric tons (550 tons)
Anticipated daily refuse volume in 3 years:
680 metric tons (750 tons)
Type of refuse: household, commercial, large amount of brush and building demolition debris
Land availability: limited, very expensive
Available Refuse Volume: 3 249 125 m^3 (4,250,000 yd^3)
Operation: 8 hours/days, 5½ days/week
Present equipment: D8 (3 years old)

What would your comments and recommendations be on the following:

a. What range of in-place densities could be expected using a track-type tractor; a Cat steel wheeled landfill compactor?
b. What effect does machine selection have on site life?
c. What are the advantages and limitations of steel wheeled landfill compactors?
d. What are the advantages and limitations of track-type units?
e. How many machines should be used on the site?
f. What type should they be?
g. What size should they be?

Solution

a. The Track-Type Tractor will achieve 475 to 595 kg/m³ (800 to 1000 lb/yd³) in-place density. The Cat steel wheeled landfill compactor will achieve 595 to 830 kg/m³ (1000 to 1400 lb/yd³) in-place density.

b. There are 3 249 125 m³ (4,250,000 yd³) available. 500 metric tons (550 tons) per day is how many m³ (yd³)? Assume a minimum density of 475 kg/m³ (800 lb/yd³).

$$500 \text{ metric tons/day} \times \frac{1000 \text{ kg/metric ton}}{475 \text{ kg/m}^3} = 1052 \text{ m}^3/\text{day}$$

$$550 \text{ tons/day} \times \frac{2000 \text{ lb/ton}}{800 \text{ lb/yd}^3} = 1375 \text{ yd}^3/\text{day}$$

5.5 days/week × 52 weeks/year = 286 days/year

Yearly volume: 1052 × 286 = 300 872 m³
1375 × 286 = 393,250 yd³

Landfill life at this density:

$$\frac{3\,250\,000 \text{ m}^3}{300\,872 \text{ m}^3/\text{year}} = \frac{4{,}250{,}000 \text{ yd}^3}{393{,}250 \text{ yd}^3/\text{year}} = 10.8 \text{ years}$$

Similar calculations are performed to generate the following tables.

500 METRIC TONS/DAY (550 TONS/DAY)

Density kg/m³	Density lb/yd³	Landfill Life (years)
475	**800**	10.8
595	**1000**	13.5
715	**1200**	16.2
835	**1400**	18.9
950	**1600**	21.6

680 METRIC TONS/DAY (750 TONS/DAY)

Density kg/m³	Density lb/yd³	Landfill Life (years)
475	**800**	7.9
595	**1000**	9.9
715	**1200**	11.9
835	**1400**	13.9
950	**1600**	15.9

From the tables we determine that a track-type tractor, at 500 metric tons per day (550 tons/day), will provide 13.5 landfill life years at 595 kg/m³ (1000 lb/yd³). Compaction will extend that life 5.4 years to 18.9 years at 835 kg/m³ (1400 lb/yd³).

Proper compaction techniques are necessary to achieve the higher refuse densities and increase landfill life.

c. Advantages: Provides highest compaction densities extending landfill life.
Limitations: Specialty unit designed to spread and compact — does not excavate virgin material economically, but can handle stockpile cover material.

d. Advantages: most versatile unit, well suited to site preparation, finishing and access road construction and maintenance; all weather machines with excellent tractive ability.
Limitation: compaction — cannot achieve the in-place refuse densities of the specialized landfill compactors.

e. Minimum of two. Additional equipment would depend on supplemental tasks.

f. Track-type tractor — for earthmoving and refuse spreading work; steel wheeled compactor-quantity of refuse and land cost would justify.

g. D8 — keeping existing unit; D9 — when new tractor is necessary; 826H — with large amount of demolition debris and brush and projected increase in tonnage would justify 826H over 816F2.

NOTE: Ballasting the wheels on Cat Landfill Compactors to increase machine weight and achieve higher compaction densities is not recommended. Landfills are high rimpull applications. Ballasting the wheels will significantly increase machine weight but decrease overall performance when traveling on the fill. The additional weight of ballasted wheels will also result in reduced final drive life.

GREEN WASTE/COMPOSTING

Green waste and compostable material are being eliminated from most landfills. Composting is normally regulated at the state level in the U.S., and on higher governmental levels within the EU. Green and compostable waste is realized as: soil amendments, bio fuels, and incinerated energy sources. With more and more green waste and compostable material being diverted from the waste stream, tools and machines that can handle this type of waste are needed. Most commonly used are wheel loaders and excavators (wheeled/tracked) of all sizes. Most are equipped with coupler for versatility of different tools. Work tools utilized range from grapple buckets, multi-purpose buckets, and light material buckets for wheel loaders, to waste grapples, orange peel grapples, and sorting grabs for excavators. It is normal, in some areas, for regulations to request added cab filtration and protection from composting airborne debris.

WASTE RELATED ENERGY PRODUCTION

There are a variety of ways to produce energy from waste. C&D landfills, green waste, biodigesters, and composting facilities, will process wood, organics and other products for: gasification, bio fuels, possible incineration, and energy from newer technologies.

LANDFILLS — RENEWABLE POWER WITH LANDFILL GAS

Landfill gas, composed mostly of methane and carbon dioxide, is produced naturally as organic waste decomposes in landfills. Driven by mandate of carbon reduction goals, modern sanitary landfills capture this gas for use as a renewable fuel in specially configured Cat landfill gas generator sets. These systems deliver reliable and environmentally sound electricity to the local community from 400 kW to 2 mW power nodes.

Please contact your local dealership, Waste representative, or *Cat.com* for more information.